

updating info sequentially

3. $P_n(H) = P'_n(H)$ (order invariance)

here $P_0(H) := P(H)$

$$P_1(H) := P(H|E_1) = \frac{P(H)P(E_1|H)}{P(E_1)}$$

$$P_2(H) := \frac{P(H)P(E_1|H)}{P(E_1)} = \frac{P(H)P(E_1|H)P(E_2|H)}{P(E_1)P(E_2)}$$

$$P_3(H) := \frac{P(H)P(E_1|H)}{P(E_1)} = \frac{P(H)P(E_1|H) \cdots P(E_n|H)}{P(E_1) \cdots P(E_n)}$$

$$P_n(H) := \frac{P_n(H)P(E_n|H)}{P(E_n)} = \frac{P(H)P(E_1|H) \cdots P(E_n|H)}{P(E_1) \cdots P(E_n)}$$

$P'_n(H)$ is obtained by E_1, E_2, E_n ,

$1', 2', \dots, n'$ is permutation of $1, 2, \dots, n$.

$$\text{proof: } P_n(H) = \frac{P(H)P(E_1|H)P(E_2|H)}{P(E_1)P(E_2)} = \frac{P(H)P(E_{n'}|H)P(E_{n'-1}|H)}{P(E_{n'})P(E_{n'-1})}$$

$$4. P(H|E_1 \cdots E_n) = P_n(H)$$

if $P(E_1 \cdots E_n) = P(E_1) \cdots P(E_n)$, $P(E_1 \cdots E_n) = P(E_1|H) \cdots P(E_n|H)$

$$\text{proof: } P(H|E_1 \cdots E_n) = \frac{P(H)P(E_1 \cdots E_n)}{P(E_1 \cdots E_n)} = \frac{P(H)P(E_1|H)}{P(E_1)}$$

$$= \frac{P(H)P(E_1|H)P(E_2|H) \cdots P(E_n|H)}{P(E_1)P(E_2) \cdots P(E_n)}$$

$$= P_n(H)$$

□

$$P(E_1)P(E_2) = P(E_1E_2)$$

$$P(E_3)P(E_1E_2) = P(E_1E_2E_3)$$

but $P(E_1)P(E_3) \neq P(E_1E_3)$

$$P(E_2)P(E_3) \neq P(E_2E_3)$$

$$P^{(0)}(H) = P(H)$$

$$P^{(1)}(H) = P^{(0)}(H|E_1)$$

$$P^{(2)}(H) = P^{(1)}(H|E_2)$$

$$= \frac{P^{(0)}(H)P(E_2|H)}{P(E_2)}$$

$$= \frac{P^{(0)}(H)P(E_2|H)}{P^{(0)}(E_2|E_1)} \cdot \frac{P^{(0)}(E_1)}{P^{(0)}(E_1)}$$

$$= P_n(H)$$

$$= \frac{P^{(0)}(H)P(E_2|H)}{P^{(0)}(E_2)}$$

$$= P^{(0)}(H|E_1E_2)$$

$$P^{(3)}(H) := P^{(2)}(H|E_3) = \frac{P^{(0)}(H)P(E_3|H)}{P^{(0)}(E_3|E_1E_2)}$$

$$= P^{(0)}(H|E_1E_2E_3)$$

...

$$P^{(n)}(H) := P^{(n-1)}(H|E_n)$$

$$= P^{(0)}(H|E_1E_2 \cdots E_n) = P(H|E_1E_2 \cdots E_n)$$

1. E_2, E_1 indep,

E_3, E_1E_2 indep

...
 $E_n, E_1 \cdots E_{n-1}E_2$ indep

$$\Rightarrow P(E_1E_2 \cdots E_n) = P(E_1)P(E_2) \cdots P(E_n)$$

proof: $P(E_1E_2E_3)$

$$= P(E_1E_2E_3)P(E_4) \quad (': E_1, E_2, E_3 \text{ indep})$$

$$= P(E_1 \cdots E_{n-1})P(E_n) \quad (': E_1, E_2, E_3 \text{ indep})$$

$$\cdots$$

$$= P(E_1)P(E_2) \cdots P(E_n) \quad (': E_2, E_1 \text{ indep})$$

1.3 E_1, E_2 indep, $\nexists \nrightarrow E_1, E_2, E_3$ indep
 E_3, E_1E_2 indep

example: $\{1, 2, \dots, 8\}$, each probability $\frac{1}{8}$.

$E_1 = \{1, 2, 3, 4\}$, $E_2 = \{1, 2, 5, 6\}$, $E_3 = \{2, 5, 6, 7\}$

$': E_1E_2 = \{1, 2\}$, $E_1E_3 = \{2\}$, $E_2E_3 = \{2, 5, 6\}$

$E_1E_2E_3 = \{2\}$.